t3-w02 — condition C3

Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

> C2 plus a validation and repair pass over the draft.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	**present**

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

- identify the types of plane shapes and state their properties, similarities and differences
- identify the parts of a circle
- find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

entry_behaviour

- Students have been taught angles.

instructional_materials

- Cardboard containing angles presentation.

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

objective_text

identify the types of plane shapes and state their properties, similarities and differences

current_step

1

instructional_events

- present_content
- elicit_performance
- present_content

assessment_points

- How many triangles has an octagon.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Hints\n\nPlane shapes are also known as two dimensional shapes (2-D shapes).
They include: triangle,\nrectangle, squares, parallelogram, trapezium, circle and other polygons.
\n\n#### Triangles\n\nIsosceles: a single tick on each of the two slanted sides, an arc at each
base angle. Scalene:\none, two and three ticks on the three sides, an arc at each vertex.
Equilateral: a single tick\non each of the three sides, an arc at each vertex.\n\n[drawing
attached to this section here, unlabelled, marked Isosceles, Scalene, Equilateral]\n\n| Isosceles
| Scalene | Equilateral |\n|---|---|---|\n| (i) 2 adjacent side are equal | No side equal | 3
sides are equal |\n| (ii) base angles are equal | No angle is equal to the other | All angles are
equal ($60^\\circ$ each) |\n| (iii) one line of symmetry | No line of symmetry. | 3 lines of
symmetry. |\n\n#### Squares and Rectangles\n\nSquare: a right-angle mark at each corner, a single
tick on each of the four sides. Rectangle: a\nright-angle mark at each corner, a single tick on
each of the two longer sides and a double tick\non each of the two shorter sides.\n\n[drawing
attached to this section here, unlabelled, marked Squares, Rectangles]\n\n| Squares | Rectangles
|\n|---|---|\n| i) All sides are equal | 2 opposite sides are equal |\n| (ii) 2 pairs of parallel
sides | 2 pair of parallel sides. |\n| (iii) All angles are equal ($90^\\circ$) each. | All
angles are equal ($90^\\circ$) each. |\n| (iv) Four lines of symmetry | Two lines of symmetry |
\n| (v) Diagonals bisect it at right angle. | Diagonals bisect it but not at right angle. |
\n\n#### Parallelograms and Trapezium\n\nParallelogram: a double arrow on each of the two
horizontal sides, a single arrow on each of the\ntwo slanted sides. Trapezium: a single arrow on
each of the two parallel sides, and one diagonal\ndrawn in.\n\n[drawing attached to this section
here, unlabelled, marked Parallelogram, Trapezium]\n\n| Parallelogram | Trapezium |\n|---|---|\n|
i) 2 pairs of opposite side are equal and parallel | Only a pair of sides are parallel |\n| (ii)
opposite angles are equal | opposite angles are not equal |\n| (iii) No line of symmetry | No
line of symmetry |\n| (iv) Diagonals divide it into equal triangles | Diagonal divide it into two
unequal triangles. |\n\n#### Polygons\n\nA polygon is a closed shape that has three or more
straight sides.\n\n| Name of Polygon | No of sides | No of triangle ($n-2$) |\n|---|---|---|\n|
Triangle | 3 | 1 |\n| Quadrilateral | 4 | 2 |\n| Pentagon | 5 | 3 |\n| Hexagon | 6 | 4 |\n|
Heptagon | 7 | 5 |\n| Octagon | 8 | 6 |\n| Nonagon | 9 | 7 |\n| Decagon | 10 | 8 |\n| Hendecagon
or undecagon | 11 | 9 |\n| Dodecagon | 12 | 10 |"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

`current_step`

2

`instructional_events`

• present_content

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Parts of a circle\n\nTwo labelled circles. The first has the diameter drawn
across it and a radius from its centre to\nthe circumference, with the arc and the circumference
labelled. The second has a chord near the\ntop with the segment above it shaded, a shaded sector
below with its right-hand bounding line\nlabelled Radius, and a tangent touching it on the right.
\n\n[drawing attached to this section here, unlabelled, marked Diameter, Radius, arc,
Circumference, Chord, Segment, Sector, Tangent]\n\n- **Arc:** This is a curved part of the
circumference.\n- **Radius:** This is any straight line joining the centre of the circle to the
circumference.\n- **Chord:** This is any straight line joining two point on the circumference of
a circle.\n- **Diameter:** Any chord that passes through the centre of the circle."
 }
]
```


Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

objective_text

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

current_step

3

assessment_points

• Find the perimeter of a square of sides 8cm
• Find the perimeter of a rectangle of length 7cm and breadth 4cm.
• What is the perimeter of regular pentagon with sides 5cm.
• Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
• The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
• Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
• Find the perimeter of a regular hexagon with sides 5cm.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Perimeter\n\nPerimeter is the distance round a shape.\n\n**Worked
examples**\n\n##### 1. Calculate the perimeter of a square of side 15cm.\n\n[drawing attached to
this section here, unlabelled, marked Square, L 15cm, 15cm, L]\n\n**Solution**\n\nPerimeter of a
square $= L + L + L + L = 4L$\n\n$= 15 + 15 + 15 + 15 = 4(15)$\n\n$= 60\\mathrm{cm}$\n\n##### 2.
Find the perimeter of a rectangle whose length 10cm and breadth 7cm.\n\n[drawing attached to this
section here, unlabelled, marked L 10cm, L, b, 7cm]\n\n**Solution**\n\nPerimeter of rectangle $=
L + b + L + b$\n\n$= L + L + b + b$\n\n$= 2L + 2b$\n\n$= 2(L + B)$\n\n$= 10\\mathrm{cm} + 7\
\mathrm{cm} + 10\\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 10\\mathrm{cm} + 10\\mathrm{cm} + 7\
\mathrm{cm} + 7\\mathrm{cm}$\n\n$= 2(10) + 2(7)$\n\n$= 2(10 + 7)$\n\n$= 2(17)$\n\n$= 34\
\mathrm{cm}$\n\n##### 3. If the perimeter of a rectangle is 60cm and its length is 17cm.
Calculate its breadth.\n\n[drawing attached to this section here, unlabelled, marked L = 17 cm,
b]\n\n**Solution**\n\nPerimeter $= 60\\mathrm{cm}$\n\n$P = L + b + L + b$\n\n$60 = 17\\mathrm{cm}
+ b + 17\\mathrm{cm} + b$\n\n$60 = 17\\mathrm{cm} + 17\\mathrm{cm} + b + b$\n\n$60\\mathrm{cm} =
2(17\\mathrm{cm}) + 2b$\n\n$60\\mathrm{cm} = 2(17\\mathrm{cm} + b)$\n\n$60\\mathrm{cm} = 34\
\mathrm{cm} + 2b$\n\ncollect like terms.\n\n$60\\mathrm{cm} - 34\\mathrm{cm} = 2b$\n\n$2b = 26\
\mathrm{cm}$\n\nDivide both side by 2\n\n$\\frac{2b}{2} = \\frac{26\\mathrm{cm}}{2}$\n\n$b = 13\
\mathrm{cm}$\n\n##### 4. Find the perimeter of the trapezium below.\n\n[drawing attached to this
section here, unlabelled, marked a 11cm, c 5cm, 9cm, b, 4cm, d]\n\nPerimeter of trapezium $= a +
b + c + d$\n\n$= 11\\mathrm{cm} + 9\\mathrm{cm} + 5\\mathrm{cm} + 4\\mathrm{cm}$\n\n$= 29\
\mathrm{cm}$\n\n##### 5. Find the perimeter of the parallelogram below.\n\n[drawing attached to
this section here, unlabelled, marked 9cm a, a, b, 5cm]\n\n**Solution**\n\nPerimeter of
parallelogram $= a + a + b + b$\n\n$= 2a + 2b$\n\n$= 2(a + b)$\n\n$= 2(9 + 5)$\n\n$= 2(14)$\n\n$=
28\\mathrm{cm}$\n\n##### 6. Find the perimeter of a regular hexagon sides 5.2cm.
\n\n**Solution.**\n\n[drawing attached to this section here, unlabelled, marked Regular hexagon,
5.2cm]\n\nHexagon has 6 sides.\n\nPerimeter $= 6 \\times 5.2\\mathrm{cm}$\n\n$= 31.2\\mathrm{cm}$
\n\n##### 7. Calculate the perimeter of circle of radius 7cm (use the value $3\\frac{1}{7}$ for $
\\pi$)\n\n**Solution**\n\n[drawing attached to this section here, unlabelled, marked 7cm]
\n\nPerimeter of a circle $= 2\\pi r$ or $\\pi d$\n\n$= 2 \\times 3\\frac{1}{7} \\times 7\
\mathrm{cm}$\n\n$= 2 \\times \\frac{22}{7} \\times 7$\n\n$= 2 \\times 22 = 44\\mathrm{cm}$
\n\nusing $\\pi d$.\n\nSince $r = 7\\mathrm{cm}$\n\n$d = 2 \\times r = 2 \\times 7\\mathrm{cm} =
14\\mathrm{cm}$\n\n$\\frac{22}{7} \\times 14\\mathrm{cm}$\n\n$= 22 \\times 2$\n\n$= 44\
\mathrm{cm}$\n\n##### 8. A rope is wound 50 times round a cylinder of radius 25cm. How long is
```

```
the rope? $\\pi = 3.14$.\n\n**Solution**\n\n[drawing attached to this section here, unlabelled]
\n\nlength of the rope $=$ circumference.\n\nCircumference $= 2\\pi r$\n\n$= 2 \\times 3.14 \
\times 25$\n\n$= 157\\mathrm{cm}$\n\nTotal length of the rope $= 157\\mathrm{cm} \\times 50$
\n\n$= 7850\\mathrm{cm}$.\n\n##### 9. The circumference of a circle is 440cm. Find the radius $(\
\pi = \\frac{22}{7})$\n\n**Solution**\n\n[drawing attached to this section here, unlabelled,
marked r]\n\nCircumference $= 2\\pi r$\n\n$440 = 2 \\times \\frac{22}{7} \\times r$\n\n$440 = \
\frac{44r}{7}$\n\ncross multiply.\n\n$440 \\times 7 = 44r$\n\n$3080 = 44r$\n\ndivide both side by
$44$\n\n$\\frac{44r}{44} = \\frac{3080}{44}$\n\n$r = 70\\mathrm{cm}$\n\n##### 10. Calculate the
perimeter of the shape below.\n\n[drawing attached to this section here, unlabelled, marked 4.9
cm, 4.9cm]\n\n**Solution**\n\nlength of the curved part $= \\frac{1}{4} \\times 2\\pi r$\n\n$= \
\frac{1}{4} \\times 2 \\times \\frac{22}{7} \\times 4.9$\n\n$= \\frac{1}{4} \\times \\frac{44}{7}
\\times 4.9$\n\n$= \\frac{11}{7} \\times \\frac{49}{10}$\n\n$= \\frac{11 \\times 7}{10}$\n\n$= \
\frac{77}{10}$\n\n$= 7.7\\mathrm{cm}$\n\nPerimeter of the shape $= 4.9 + 4.9 + 7.7$\n\n$= 17.5\
\mathrm{cm}$"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Types of Plane Shapes and Their Properties

EXPLANATION

A plane (2-D) shape is a flat figure that lies in a single plane. The main shapes you need to know are:

Triangles – three sides. They come in three kinds:
- *Isosceles*: two sides equal, the angles at the base equal, one line of symmetry.
- *Scalene*: no sides equal, no angles equal, no symmetry.
- *Equilateral*: all three sides equal, all angles 60°, three lines of symmetry.

Quadrilaterals – four sides. Important types are:
- *Square*: all sides equal, all right angles, four lines of symmetry, diagonals bisect each other at right angles.
- *Rectangle*: opposite sides equal, all right angles, two lines of symmetry, diagonals bisect each other (but not at right angles).
- *Parallelogram*: opposite sides equal and parallel, opposite angles equal, no lines of symmetry, diagonals bisect each other.
- *Trapezium*: only one pair of sides parallel, no symmetry, diagonals divide the shape into two unequal triangles.

Other polygons – shapes with more than four sides. They are named by the number of sides (pentagon, hexagon, …). A useful fact is that any polygon with *n* sides can be divided into *(n-2)* triangles.

These descriptions let you compare shapes: look at side lengths, angle measures, parallelism, and symmetry.

Remember: the tick marks, arrows and arcs you may see in diagrams (fig-01, fig-02, fig-03) are just visual clues for these properties.

WORKED_EXAMPLE

Worked example

Identify the shape shown below (imagine the diagram from fig-02): it has four right-angle marks at each corner, a single tick on each side, and all sides are the same length.

Solution: The right-angle marks tell us every interior angle is 90°, and the equal ticks on all four sides tell us the sides are equal. That matches the definition of a **square**. Therefore the shape is a square.

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(4,0)--(2,4)--cycle;
  \draw (0.8,1.6)--(1.2,1.4);
  \draw (3.2,1.6)--(2.8,1.4);
  \draw (0,0)++(0:0.7) arc[start angle=0, end angle=63, radius=0.7];
  \draw (4,0)++(180:0.7) arc[start angle=180, end angle=117, radius=0.7];
  \node[below] at (2,-0.6) {Isosceles};
  \begin{scope}[xshift=7cm]
    \draw (0,0)--(5,0)--(1.4,3.4)--cycle;
    \draw (0.38,1.83)--(1.02,1.57);
    \draw (3.02,1.87)--(3.50,2.38);
    \draw (3.38,1.53)--(3.86,2.04);
    \foreach \d in {-0.25,0,0.25} { \draw (2.5+\d,-0.25)--(2.5+\d,0.25); }
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=68, radius=0.7];
    \draw (5,0)++(180:0.8) arc[start angle=180, end angle=137, radius=0.8];
    \draw (1.4,3.4)++(-113:0.7) arc[start angle=-113, end angle=-25, radius=0.7];
    \node[below] at (2.5,-0.6) {Scalene};
  \end{scope}
  \begin{scope}[xshift=14cm]
    \draw (0,0)--(4,0)--(2,3.46)--cycle;
    \draw (0.8,1.55)--(1.2,1.32);
    \draw (3.2,1.55)--(2.8,1.32);
    \draw (1.8,-0.25)--(2.2,0.25);
    \draw (0,0)++(0:0.7) arc[start angle=0, end angle=60, radius=0.7];
    \draw (4,0)++(180:0.7) arc[start angle=180, end angle=120, radius=0.7];
    \draw (2,3.46)++(-120:0.7) arc[start angle=-120, end angle=-60, radius=0.7];
```

```
    \node[below] at (2,-0.6) {Equilateral};
  \end{scope}
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (3.4,3.4);
  \draw (0,0.5)--(0.5,0.5)--(0.5,0);
  \draw (3.4,0.5)--(2.9,0.5)--(2.9,0);
  \draw (0,2.9)--(0.5,2.9)--(0.5,3.4);
  \draw (3.4,2.9)--(2.9,2.9)--(2.9,3.4);
  \draw (1.7,-0.25)--(1.7,0.25);
  \draw (1.7,3.15)--(1.7,3.65);
  \draw (-0.25,1.7)--(0.25,1.7);
  \draw (3.15,1.7)--(3.65,1.7);
  \node[below] at (1.7,-0.9) {Squares};
  \begin{scope}[xshift=8cm]
    \draw (0,0) rectangle (5,3);
    \draw (0,0.5)--(0.5,0.5)--(0.5,0);
    \draw (5,0.5)--(4.5,0.5)--(4.5,0);
    \draw (0,2.5)--(0.5,2.5)--(0.5,3);
    \draw (5,2.5)--(4.5,2.5)--(4.5,3);
    \draw (2.5,-0.25)--(2.5,0.25);
    \draw (2.5,2.75)--(2.5,3.25);
    \draw (-0.25,1.35)--(0.25,1.35); \draw (-0.25,1.65)--(0.25,1.65);
    \draw (4.75,1.35)--(5.25,1.35); \draw (4.75,1.65)--(5.25,1.65);
    \node[below] at (2.5,-0.9) {Rectangles};
  \end{scope}
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(5,0)--(6.6,3)--(1.6,3)--cycle;
  \draw[->] (2.4,0)--(2.6,0);
  \draw[->] (3.9,3)--(4.1,3);
  \draw[->] (5.7,1.4)--(5.85,1.65);
  \draw[->] (5.85,1.4)--(6.0,1.65);
  \draw[->] (0.7,1.4)--(0.85,1.65);
  \draw[->] (0.85,1.4)--(1.0,1.65);
  \node[below] at (3,-0.6) {Parallelogram};
  \begin{scope}[xshift=10cm]
    \draw (0,0)--(6,0)--(4.6,3)--(1.6,3)--cycle;
    \draw (0,0)--(4.6,3);
    \draw[->] (2.9,0)--(3.1,0);
    \draw[->] (3.0,3)--(3.2,3);
    \node[below] at (3,-0.6) {Trapezium};
  \end{scope}
\end{tikzpicture}
```

1. 1. A shape has three equal sides and three equal angles. What type of triangle is it?
answer: Equilateral triangle

2. 2. A quadrilateral has opposite sides equal and parallel, but none of its angles are right angles. Which shape is it?
answer: Parallelogram

3. 3. How many triangles can you divide a regular hexagon into?
answer: 4

4. 4. A quadrilateral has only one pair of opposite sides parallel. Which shape is it?
answer: Trapezium

5. 5. Which quadrilateral has four lines of symmetry?
answer: Square

Parts of a Circle

EXPLANATION
A circle is made up of several important parts:

- **Circumference** – the complete outer edge of the circle.
- **Arc** – any curved portion of the circumference.
- **Radius** – a straight line from the centre of the circle to any point on the circumference.
- **Diameter** – a special chord that passes through the centre; it is twice the length of a radius.
- **Chord** – a straight line joining two points on the circumference (the diameter is a type of chord).
- **Sector** – the region bounded by two radii and the arc between them.
- **Segment** – the region bounded by a chord and the arc subtended by the chord.
- **Tangent** – a straight line that touches the circle at exactly one point.

These parts are shown in figure fig-04. Notice how the radius meets the centre, the diameter stretches across the whole circle, and the chord lies entirely inside the circle and does not pass through the centre (unless it is a diameter).

VISUAL
```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (-3,0)--(3,0);
  \node[above] at (0,0.1) {Diameter};
  \draw (0,0)--(0,-3);
  \node[right] at (0.45,-1.9) {Radius};
  \draw[->] (-5.9,-3.4)--(-2.3,-1.9);
  \node[left] at (-5.9,-3.4) {arc};
  \draw[->] (4.4,3.4)--(2.2,2.2);
  \node[right] at (4.4,3.4) {Circumference};
  \begin{scope}[xshift=17cm]
    \draw (0,0) circle (3);
    \draw (-2.29,1.94)--(2.29,1.94);
    \begin{scope}
      \clip (0,0) circle (3);
      \clip (-3,1.94) rectangle (3,3);
      \foreach \x in {-3,-2.7,...,3} { \draw (\x,1.8)--(\x+1.2,3.0); }
    \end{scope}
    \node[below] at (0,1.9) {Chord};
    \draw[->] (3.2,4.2)--(0.6,2.5);
    \node[right] at (3.2,4.2) {Segment};
    \begin{scope}
      \clip (0,0)--(-80:3) arc[start angle=-80, end angle=-25, radius=3]--cycle;
      \foreach \y in {-3.4,-3.1,...,0} { \draw (0,\y)--(3,\y+1.2); }
    \end{scope}
    \draw (0,0)--(-80:3);
    \draw (0,0)--(-25:3) node[midway, sloped, above] {Radius};
    \draw[->] (-3.6,-4.2)--(1.0,-2.0);
    \node[left] at (-3.6,-4.2) {Sector};
    \draw (3,-2.4)--(3,2.4);
    \node[right] at (3.2,0) {Tangent};
  \end{scope}
\end{tikzpicture}
```

WORKED_EXAMPLE
Worked Example:

In figure fig-04 the left-hand circle shows a diameter drawn across the centre and a radius drawn from the centre to the edge. The arc and the whole circumference are also labelled. In the right-hand circle you can see:
- a **chord** near the top,
- a **sector** shaded below the chord,

- a **radius** drawn from the centre to the edge of the sector,
- a **tangent** touching the circle on the right side.

From this picture you can identify each part:
- The straight line that goes through the centre is the **diameter**.
- Any line from the centre to the edge, like the one on the right, is a **radius**.
- The curved line around the whole shape is the **circumference**; a piece of it is an **arc**.
- The line joining the two top points is a **chord**.
- The shaded wedge is a **sector**, and the area between the chord and the arc is a **segment**.
- The line that just touches the circle on the right is a **tangent**.

Understanding these parts helps you describe and work with circles in geometry problems.

1. 1. In a circle, what is the name of the line that joins the centre to any point on the circumference?
answer: Radius

2. 2. What do you call a straight line that joins two points on the circumference but does not pass through the centre?
answer: Chord

3. 3. Identify the part of the circle that is a curved portion of the circumference.
answer: Arc

4. 4. Which part of a circle is a straight line that touches the circle at exactly one point?
answer: Tangent

5. 5. If a line passes through the centre and its endpoints lie on the circle, what is this line called?
answer: Diameter

6. 6. The region bounded by two radii and the arc between them is known as a ____.
answer: Sector

7. 7. The part of the circle that is the whole outer edge is called the ____.
answer: Circumference

Finding Perimeters of Common Plane Shapes

EXPLANATION

The perimeter of a shape is the total distance around its edge. For regular polygons and other familiar shapes you can use simple formulas:

- Square: $P = 4 \times \text{side}$
- Rectangle: $P = 2(\text{length} + \text{breadth})$
- Regular n-gon: $P = n \times \text{side}$
- Parallelogram: $P = 2(\text{base} + \text{side})$
- Trapezium: $P = \text{side}_1 + \text{side}_2 + \text{side}_3 + \text{side}_4$
- Circle: $C = 2\pi r = \pi d$ (called the circumference)

Remember to keep the units consistent (centimetres, metres, etc.) and write your answer with the appropriate unit.

WORKED_EXAMPLE

Example 1 – Square

A square has each side equal to 15 cm (see fig-05). Its perimeter is:
$$P = 4 \times 15 = 60\text{ cm}$$

Example 2 – Rectangle

For a rectangle with length 10 cm and breadth 7 cm (see fig-06):
$$P = 2(10 + 7) = 2 \times 17 = 34\text{ cm}$$

Example 3 – Finding an Unknown Side

A rectangle's perimeter is 60 cm and its length is 17 cm (see fig-07). Let the breadth be b:
$$60 = 2(17 + b) \Rightarrow 30 = 17 + b \Rightarrow b = 13\text{ cm}$$

Example 4 – Trapezium

Sides are $a = 11$ cm, $b = 9$ cm, $c = 5$ cm, $d = 4$ cm (see fig-08).
$$P = a + b + c + d = 11 + 9 + 5 + 4 = 29\text{ cm}$$

Example 5 – Parallelogram

Base $a = 9$ cm, side $b = 5$ cm (see fig-09).
$$P = 2(a + b) = 2(9 + 5) = 28\text{ cm}$$

Example 6 – Regular Hexagon

Each side is 5.2 cm (see fig-10).
$$P = 6 \times 5.2 = 31.2\text{ cm}$$

Example 7 – Circle

Radius $r = 7$ cm, use $\pi = \frac{22}{7}$ (see fig-11).
$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44\text{ cm}$$

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (5,5);
  \node[left]  at (-1.4,4.4) {Square};
  \node[above] at (2.5,5) {$L$\quad 15cm};
  \node[below] at (2.5,0) {$L$\quad 15cm};
  \node[left]  at (0,3) {15cm};
  \node[left]  at (0,1.8) {$L$};
```

```
  \node[right] at (5,3) {15cm};
  \node[right] at (5,1.8) {$L$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4.5);
  \node[above] at (3.5,4.5) {$L$\quad 10cm};
  \node[below] at (3.5,0) {$L$};
  \node[left]  at (0,2.25) {$b$};
  \node[right] at (7,2.7) {$b$};
  \node[right] at (7,1.6) {7cm};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) rectangle (7,4);
  \node[above] at (3.5,4) {$L = 17$cm};
  \node[below] at (3.5,0) {$L = 17$cm};
  \node[left]  at (0,2) {$b$};
  \node[right] at (7,2) {$b$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(7.5,0)--(6.0,3.2)--(2.0,3.2)--cycle;
  \node[below] at (3.75,0) {$a$\quad 11cm};
  \node[above] at (4.0,3.2) {$c$\quad 5cm};
  \node[left]  at (0.95,2.3) {9cm};
  \node[left]  at (0.65,1.35) {$b$};
  \node[right] at (6.9,2.3) {4cm};
  \node[right] at (7.15,1.35) {$d$};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(6,0)--(8,3.2)--(2,3.2)--cycle;
  \draw[->] (2.7,0)--(2.9,0);
  \draw[->] (4.7,3.2)--(4.9,3.2);
  \draw[->] (0.9,1.5)--(1.05,1.75);
  \draw[->] (6.9,1.5)--(7.05,1.75);
  \node[below] at (3,0) {9cm\quad $a$};
  \node[above] at (5,3.2) {$a$};
  \node[left]  at (0.9,1.6) {$b$};
  \node[right] at (7.2,1.9) {$b$};
  \node[right] at (7.4,1.1) {5cm};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \foreach \i in {0,...,5} { \coordinate (H\i) at ({\i*60}:3); }
  \draw (H0)--(H1)--(H2)--(H3)--(H4)--(H5)--cycle;
  \node[above]       at (0,4.0) {Regular hexagon};
  \node[above]       at (0,2.65) {5.2cm};
  \node[above right] at (2.4,1.3) {5.2cm};
  \node[below right] at (2.4,-1.3) {5.2cm};
  \node[below]       at (0,-2.65) {5.2cm};
  \node[below left]  at (-2.4,-1.3) {5.2cm};
  \node[above left]  at (-2.4,1.3) {5.2cm};
\end{tikzpicture}
```

VISUAL

```
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
```

```
  \draw (0,0)--(3,0) node[midway,above] {7cm};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) ellipse (2 and 0.6);
  \draw (-2,0)--(-2,-6);
  \draw (2,0)--(2,-6);
  \draw (-2,-6) arc[start angle=180, end angle=360, x radius=2, y radius=0.6];
  \foreach \y in {-0.9,-1.8,-2.7,-3.6,-4.5,-5.4} {
    \draw (-2,\y) arc[start angle=180, end angle=360, x radius=2, y radius=0.55];
    \draw (2,\y-0.03)--(-2,\y-0.9);
  }
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0) circle (3);
  \draw (0,0)--(3,0) node[midway,above] {$r$};
\end{tikzpicture}
```

```
VISUAL
\begin{tikzpicture}[scale=0.42]
  \draw (0,0)--(0,5)--(0,5) arc[start angle=90, end angle=0, radius=5]--(0,0);
  \node[left]  at (0,2.5) {4.9 cm};
  \node[below] at (2.5,0) {4.9cm};
\end{tikzpicture}
```

1. Find the perimeter of a square whose side is 8 cm.
answer: 32 cm

2. Find the perimeter of a rectangle with length 7 cm and breadth 4 cm.
answer: 22 cm

3. What is the perimeter of a regular pentagon with each side 5 cm?
answer: 25 cm

4. Find the perimeter of a parallelogram whose base is 6 cm and side is 5 cm.
answer: 22 cm

5. The circumference of a circle is 22 cm. Find the radius (use \(\pi = \frac{22}{7}\)).
answer: 1 cm

6. A rectangle's length is twice its breadth. If the breadth is \(d\) cm, write an expression for the total length of all four sides.
answer: 6d cm

7. Calculate the perimeter of a regular hexagon with each side 5 cm.
answer: 30 cm

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	3	Scope	**grounded**
Explanation the principle, rule or relationship to review	3	Enacted content	**grounded**
Worked example a problem with its solution worked through	3	Enacted content	**grounded**
Figure a drawing the learner reads from	14	Figures	**grounded**

Practice what the learner does	19	Assessment written from the teacher's evaluation questions instead of her set work	**substituted**
Answer what the learner checks against	19	Enacted content	**grounded**